



## Policy

# NIISWA and Neurosurgery - Clinical Governance of Endovascular and Neurovascular Interventions

## Scope

| Site | Service/Department/Unit | Disciplines   |
|------|-------------------------|---------------|
| SCGH | NIISwa, Neurosurgery    | Medical Staff |

## Aim

This Policy outlines the medical clinical governance structure for the interface between NIISWA and Neurosurgery with respect to neurovascular endovascular procedures conducted at Sir Charles Gairdner Hospital (SCGH) within the North Metropolitan Health Services (NMHS).

## Risk Statement

Non-compliance with this policy will:

|                                       |                                     |                                   |                                     |
|---------------------------------------|-------------------------------------|-----------------------------------|-------------------------------------|
| Breach legislative requirements       | <input type="checkbox"/>            | Impact on Patient Quality of Care | <input checked="" type="checkbox"/> |
| Breach National/State/Hospital Policy | <input checked="" type="checkbox"/> | Impact on Patient Safety          | <input checked="" type="checkbox"/> |
| Breach professional standards         | <input type="checkbox"/>            | Misconduct                        | <input type="checkbox"/>            |
| Breach SCGOPHCG Mission & Values      | <input type="checkbox"/>            | Staff Safety                      | <input type="checkbox"/>            |
| Other:                                | <input type="checkbox"/>            |                                   |                                     |

## Definitions

|                                                                                         |                                                                                                             |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Neurological Imaging and Interventional Service of Western Australia (NIISwa)           | NIISwa is a state-wide, cross campus neurointerventional and advanced neuroimaging service.                 |
| Conjoint Committee for Recognition of Training in Interventional Neuroradiology (CCINR) | CCINR develops and monitors guidelines for certification of training in interventional neuroradiology (INR) |
| FRACS                                                                                   | Fellow of the Royal Australasian College of Surgeons                                                        |
| Statewide Neurosurgical Service                                                         | Neurosurgery is a state-wide, cross campus neurosurgical service.                                           |

## Policy Principles

### 1. Clinical Governance

NIISWA and Neurosurgery are committed to a collaborative and mutually respectful and shared model of care.

All senior medical practitioners shall adhere to their defined scope of practice at all times.

**Brain aneurysms are to be treated with endovascular techniques except in cases where there is agreement between a neurointerventionist and neurosurgeon that clipping is a more appropriate treatment modality for that patient.**

### 2. Management of Endovascular Neurointerventional Procedures

NIISwa has clinical governance over endovascular neurointerventional procedures performed at SCGH. This includes:

- NIISWA will select which practitioners can perform endovascular neurointerventional procedures at SCGH and determine the appropriate level of supervision.
- Any medical practitioner performing or assisting in any endovascular neurointerventional procedure will report to the NIISwa neurointervention Lead and to the NIISwa Head of Department (HoD) with regards to:
  - The decision to treat endovascularly
  - The procedure
  - Periprocedural decision-making with regards to the procedure
  - Post-procedure follow-up with regards to the procedure
  - Clinical audit
  - Reporting of complications
  - Reporting of morbidity and mortality

NIISwa HOD has clinical governance over training and supervision in endovascular neurointerventions. This includes:

- The selection of trainees, fellows and consultants to perform endovascular neurointervention at SCGH/NMHS;
- The inclusion of endovascular neurointervention in the scope of practice of new or existing NMHS employees.
- The rostering and supervision of any endovascular neurointervention trainee or consultants.

A consultant neurosurgeon (FRACS), who is appointed to the State-wide Neurosurgical Service, with CCINR accreditation, can only be appointed collaboratively, in conjunction with NIISWA. The Neurosurgeons principle appointment is to the neurosurgical service, they can be made an honorary consultant within NIISwa, at the discretion of the NIISwa HoD for the purpose of the clinical governance structure previously outlined.

### 3. Management of Aneurysmal Subarachnoid Haemorrhage (aSAH) Patients

#### 3.1 aSAH patients

- New admission of aSAH patients must be discussed with the on-call open cerebrovascular surgeon together with the on-call NIISwa endovascular neurointerventionist, prior to a decision regarding the treatment of the aneurysm.
- All aSAH patients will be admitted under the open cerebrovascular

neurosurgeon admitting medical officer 1 (AMO1) and a NIISwa endovascular neurointerventionist admitting medical officer admitting medical officer 2 (AMO2).

- For patients with a Middle Cranial Artery (MCA) Aneurysm with clinical features of raised intracranial pressure secondary to a haematoma, a neurosurgical treatment modality should be routinely offered.
- Any endovascular treatment of aSAH patients admitted under an open cerebrovascular neurosurgeon (AMO1) who also has a second role as an honorary endovascular neurointerventionist within NIISwa, will be performed by the NIISwa endovascular neurointerventionist rostered on the relevant day, regardless of which open cerebrovascular neurosurgeon (AMO1) the patient is admitted under.
- If a practitioner is the AMO1 of an aSAH patient as part of their open cerebrovascular neurosurgeon role, and that practitioner is also an honorary endovascular consultant within NIISwa, these two roles are separate with regards to the clinical governance of endovascular neurointervention.

### 3.2 Elective Aneurysm treatment patients

- The patient will be admitted under the bed-card (AMO1) of the consultant performing the aneurysm treatment procedure (either the NIISwa endovascular neurointerventionist or open cerebrovascular neurosurgeon) unless otherwise agreed.
- The junior practitioners of the admitting consultant (either the NIISwa endovascular neurointerventionist or open cerebrovascular neurosurgeon) will attend to inpatient issues during office hours as well as after-hours.

## 4. Management of Outpatient Referrals

### 4.1 Vascular referrals

- Referrals categorised as “unruptured aneurysms” during triage will be forwarded on eReferral directly to NIISwa.

### 4.2 Neurosurgical referrals

- “Unruptured or incidental aneurysms” related neurosurgical referrals will be triaged by neurosurgery and then referred on to NIISwa for management if required.

### 4.3 Brain Aneurysm referrals

- Referring general practitioners (GPs) and regional referrers will need to send unruptured brain aneurysm referrals to NIISwa via the Central Referral Service (CRS) Health Pathways website.
  - Brain aneurysm referrals to NIISwa will continue to be managed by the NIISwa neurointervention Lead and distributed to NIISwa consultant clinics or open cerebrovascular neurosurgery consultant clinics.
- Aneurysm subtypes and patient subtypes with equipoise as to the better treatment modality will be discussed (usually following cerebral angiography) at the weekly multidisciplinary cerebrovascular Multidisciplinary Team (MDT) meeting

### 4.4 Other referrals

- Brain Arterio-Venous Malformation (AVM) referrals will be (re)directed to the open cerebrovascular neurosurgery consultant clinic routinely.
- Referrals that are sent directly to the neurosurgery department will be assessed and distributed accordingly by the neurosurgery department.

## 5. On-Call rostering for Endovascular Neurointervention

The after-hours NIISwa endovascular neurointervention roster is a state-wide service providing:

- Emergency immediate mechanical thrombectomy for stroke
- Realtime immediate availability for consultation regarding the suitability of patients for mechanical thrombectomy
- Reporting of hyperacute stroke imaging
- Endovascular treatment of ruptured brain aneurysms
- Endovascular treatment of SAH related vasospasm.
- Endovascular treatment of acute extra-cranial head and neck haemorrhage (trauma, neoplastic, and epistaxis)

The rostered on-call endovascular neurointerventionist must be appropriately qualified and experienced to perform all the above independently and without supervision.

The rostered on-call endovascular neurointerventionist cannot be on-call concurrently for any other service within or external to SCGH.

### Compliance and Monitoring

Both the NIISWA HoD and Neurosurgical HoD will be jointly responsible for monitoring compliance with this Policy. Regular audit may be conducted to evaluate the compliance with this policy.

### References

Central Referral Service - [About the Central Referral Service \(health.wa.gov.au\)](http://health.wa.gov.au)

### Authorisation & Version Control

|                               |                                                                                                                                                                                                                  |                |            |              |            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------------|------------|
| Policy Sponsor                | Area Director Clinical Services SCGOPHCG                                                                                                                                                                         |                |            |              |            |
| Policy Contact                | Consultant Neuroradiology                                                                                                                                                                                        |                |            |              |            |
| Date First Issued:            | 22/07/2022                                                                                                                                                                                                       | Last Reviewed: | 12/08/2024 | Review Date: | 12/08/2027 |
| Version No.                   | 2                                                                                                                                                                                                                | Reference #:   | HP115      |              |            |
| Approved by:                  | Area Director Clinical Services<br>SCGOPHCG                                                                                                                                                                      |                |            | Date:        | 12/08/2024 |
| Endorsed by:                  | SCGOPHCG Policy and Guideline Committee                                                                                                                                                                          |                |            | Date:        | 22/07/2022 |
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| Standards Applicable          |                                                                                                                               |                |            |              |            |
| Aboriginal Health ISD Number: | 3565                                                                                                                                                                                                             |                |            |              |            |

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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